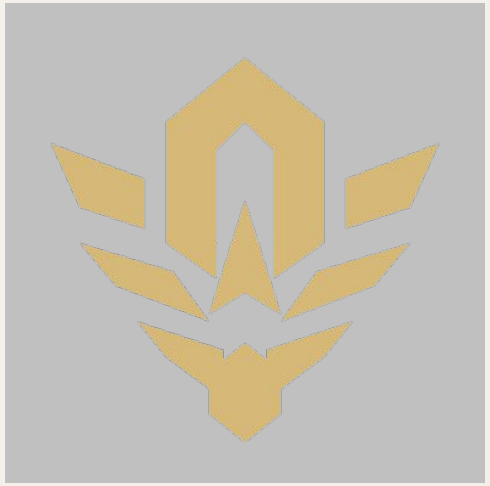
Multilayer Smart Security System

Using Sensor Fusion and ML Classification



Aegis

Motion and distance sensing with visual person verification

PROJECT TEAM

Slot: E1+TE1

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The Problem Statement

A motion trigger alone cannot verify a person.

An unnecessary alarm creates work. A missed person creates risk. The design must balance both.

Single sensor

PIR responds to changes in infrared radiation. It does not classify the object.

Dual sensing + Vision

PIR and ultrasonic form one sensing layer. Camera + ML provide a second verification layer.

Research motivation: [1] false-alarm constraints, [2] PIR–ultrasonic sensing, [3] fusion with visual detection.

Two Layer Architecture

LAYER 1

Dual sensing

PIR detects motion-related infrared changes.
Ultrasonic checks distance to the object.

Three agreements trigger camera capture and a 1-second beep. The beep works offline.

LAYER 2

Camera + ML

ESP32-CAM captures the event.
Laptop YOLO checks whether a person is visible.

A new ML person result returns over Wi-Fi
and triggers a 5-second alarm on the same buzzer.

Dual sensing means two complementary sensors within one layer. Person detection does not establish authorization.

Research Basis, Gaps and Aegis Contribution

Paper	Key contribution	Gap addressed by Aegis
[1] Pugh et al., 2015 Intrusion detection	Multiple PIR sensors with an explicit false-alarm constraint.	Combines PIR detection with ultrasonic distance gating and camera-based person verification
[2] Ciuffreda et al., 2023 Indoor localization	PIR + ultrasonic sensing, with decision-tree and clustering methods.	Applies complementary sensors to trigger camera capture and visual classification within a fixed monitoring zone
[3] Patino et al., 2022 Border surveillance	PIR + RGB + thermal detection with weighted fusion and tracking.	Implements a short-range dual-sensor trigger, ESP32-CAM person classification and local alarm feedback

Aegis combines candidate sensing, visual verification and local alarm feedback.

Key Advantages Over Existing Research

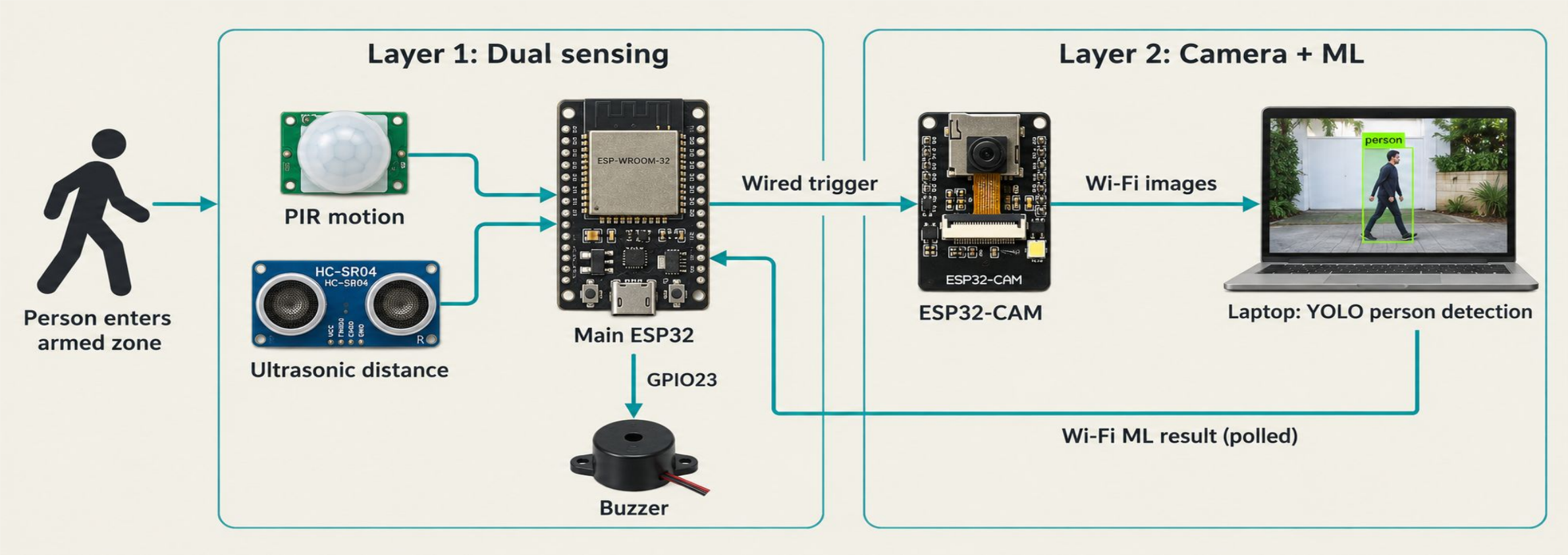
Existing approach	What Aegis adds	Advantages
PIR-based intrusion detection	PIR + ultrasonic distance gating + YOLO verification	Filters irrelevant activity through multiple verification stages
Fixed indoor surveillance networks	Compact ESP32-based hardware with a relocatable monitoring setup	Easier repositioning for different rooms and entry points
Continuous camera monitoring	Camera capture only after dual-sensor agreement	Lower camera power consumption through reduced operating time

Key Advantages: Portable, Low-cost hardware, Dual-sensor false-trigger filtering, Low power consumption

System architecture and workflow

Dual sensing: 1-second buzzer beep

ML confirmation: 5-second alarm



Main ESP32 polls the ML result over Wi-Fi. GPIO23 drives the buzzer: 1-second sensor beep, 5-second ML alarm.

Illustrative comparison of three configurations

Hypothetical example: 1,000 person events and 1,000 no-person events

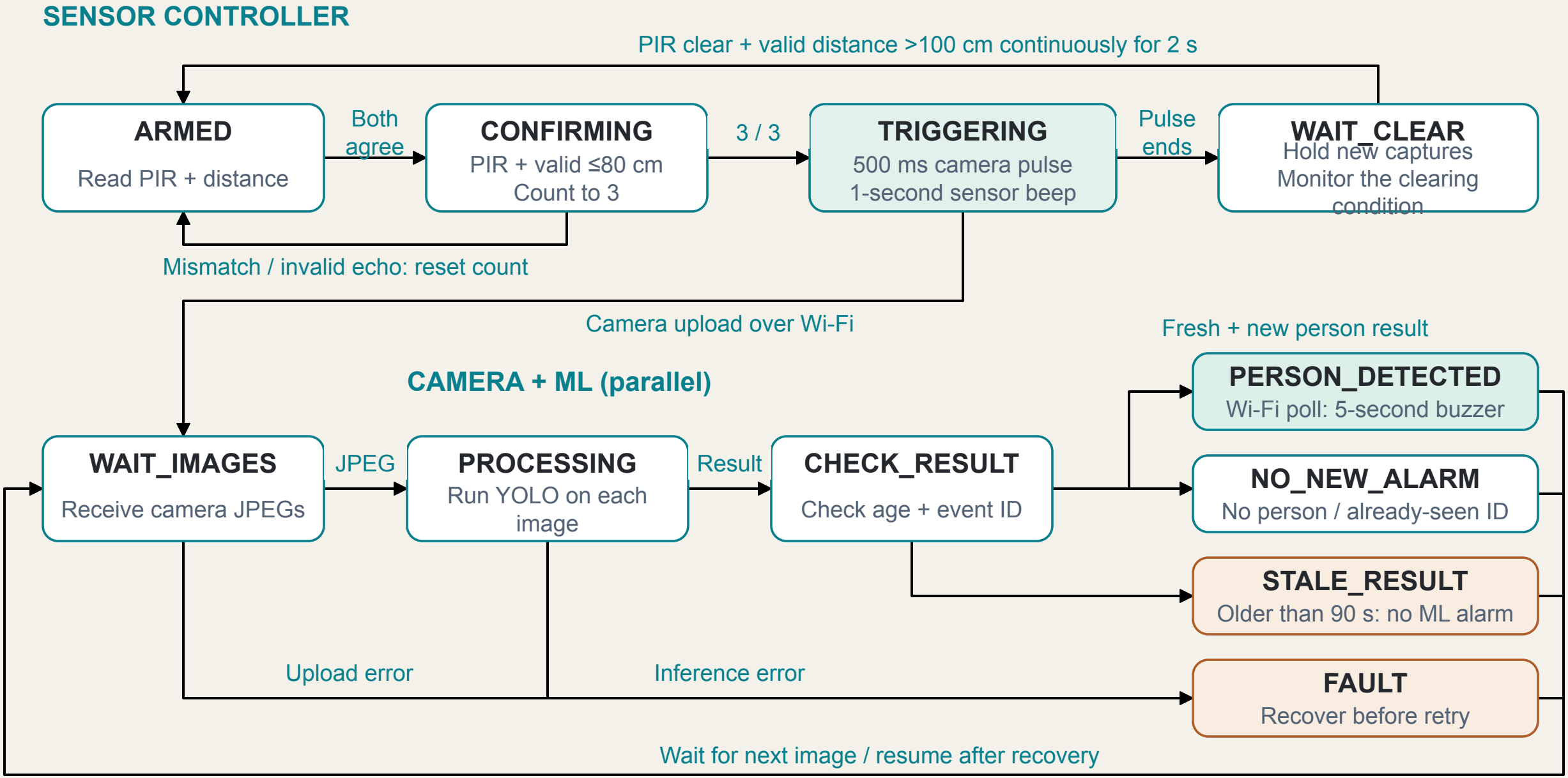
Configuration	People detected	People missed	False positives	Recall (%)	FP rate (%)
PIR only	953 / 1,000	47 / 1,000	397 / 1,000	95.3%	39.7%
PIR + ultrasonic	907 / 1,000	93 / 1,000	148 / 1,000	90.7%	14.8%
PIR + ultrasonic + camera + ML	856 / 1,000	144 / 1,000	27 / 1,000	85.6%	2.7%

Extra checks can reject nuisance events, but can also miss real people.

Fraction to percentage: 953 / 1,000 × 100 = 95.3%.

All numbers above are hypothetical teaching figures, not measured project results.
They illustrate the final ML-confirmed alarm. The 1-second sensor beep may still sound.

FSM Diagram


SENSOR CONTROLLER
PIR clear + valid distance >100 cm continuously for 2 s
ARMED
Read PIR + distance
Both agree
CONFIRMING
PIR + valid ≤80 cm
Count to 3
3 / 3
TRIGGERING
500 ms camera pulse
1-second sensor beep
Pulse ends
WAIT_CLEAR
Hold new captures
Monitor the clearing condition
Mismatch / invalid echo: reset count
Camera upload over Wi-Fi
CAMERA + ML (parallel)
Fresh + new person result
WAIT_IMAGES
Receive camera JPEGs
JPEG
PROCESSING
Run YOLO on each image
Result
CHECK_RESULT
Check age + event ID
PERSON_DETECTED
Wi-Fi poll: 5-second buzzer
NO_NEW_ALARM
No person / already-seen ID
STALE_RESULT
Older than 90 s: no ML alarm
Upload error
Inference error
FAULT
Recover before retry
Wait for next image / resume after recovery

Sensor decision table: when the first layer responds

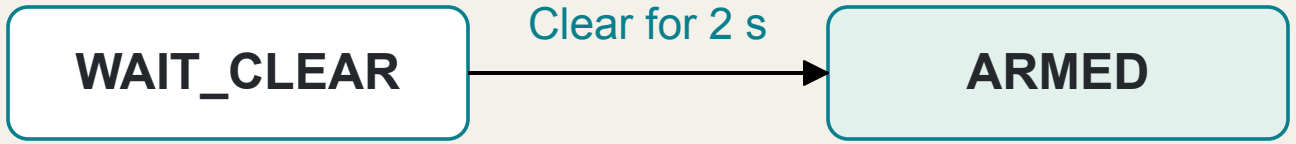
PIR state	Ultrasonic reading	Agreement count	Action while armed
No motion	Valid distance >80 cm	Reset to 0	Continue monitoring.
Motion	Valid distance >80 cm	Reset to 0	No sensor-triggered capture or beep.
No motion	Valid distance ≤80 cm	Reset to 0	Nearby object alone does not trigger.
Motion	Valid distance ≤80 cm	Add 1; requires 3 consecutive samples	At count 3: camera trigger + 1-second beep.
Either state	No echo / invalid	Reset to 0	No new sensor-triggered event.

Both sensors must agree. An invalid echo is not evidence that the area is clear.

These rules apply to automatic sensor triggering. ML feedback is a separate alarm path.

Alarm feedback and rearming

RETURN TO ARMED



All clearance conditions must hold

PIR reads CLEAR (LOW).
Valid distance remains above 100 cm.
Both stay true continuously for 2 seconds.

Otherwise, restart the clearance timer

Motion, distance ≤100 cm or an invalid echo
resets the timer and keeps WAIT_CLEAR active.

ALARM FEEDBACK

Sensor feedback

1-second beep after sensor confirmation.
Works independently of Wi-Fi.

ML feedback

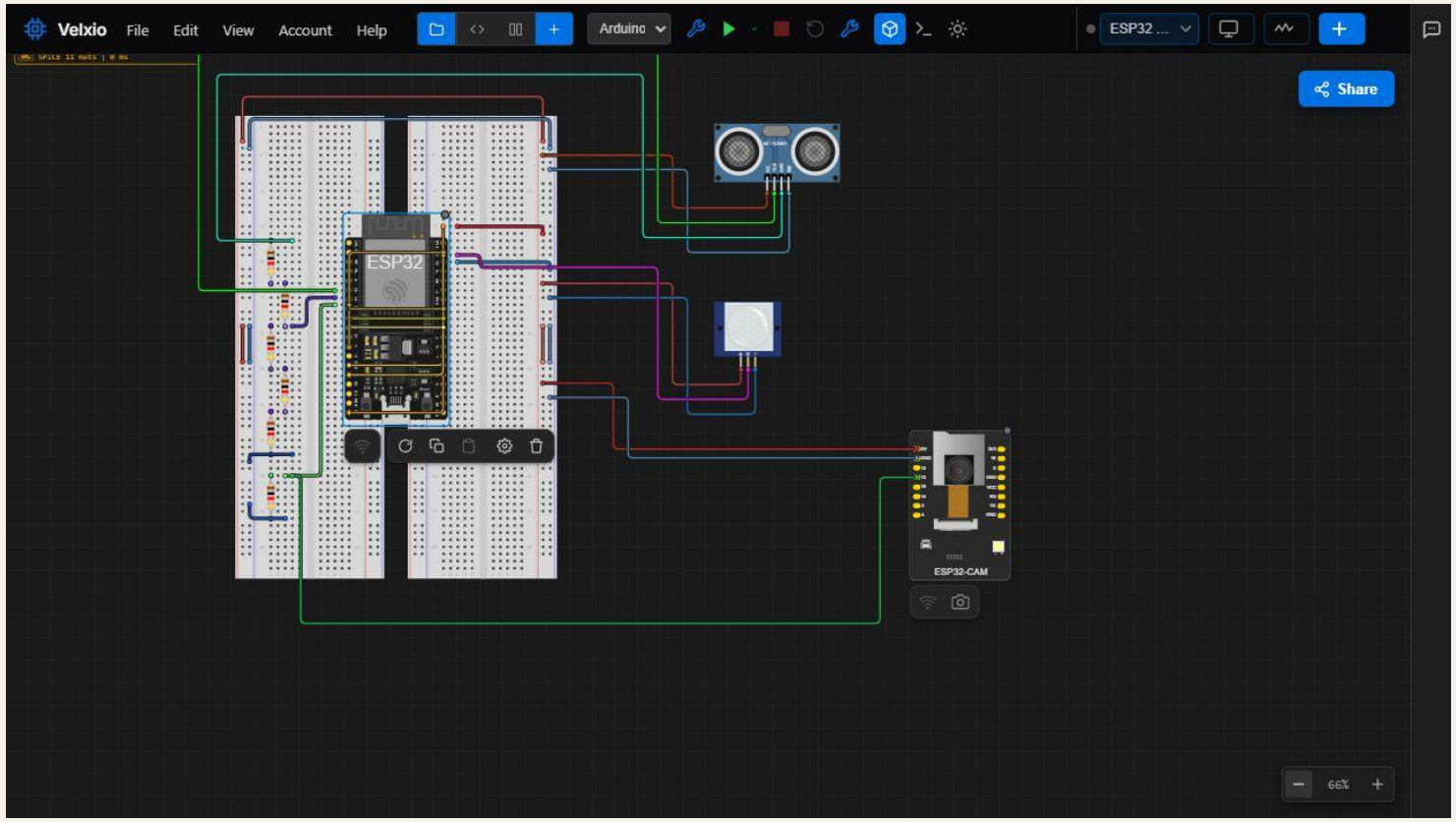
The ESP32 polls over Wi-Fi. A fresh, new
person-positive event requests a 5-second alarm.

Priority and automatic stop

The ML alarm overrides the short beep.
The buzzer stops when its timer expires.

Rearming follows sensor clearance. It does not wait for ML inference or for the buzzer to finish.

Hardware wiring and circuit reference



Signal	Connection
PIR OUT	DevKit GPIO13
HC-SR04 TRIG	DevKit GPIO5
HC-SR04 ECHO	Divider to GPIO18
Camera trigger	GPIO19 to CAM GPIO13
Passive piezo	GPIO23 + GND

Common GND. Regulated 5 V board supply.
Keep ESP32 inputs within 3.3 V logic levels.

Circuit screenshot is the earlier reference. The current physical setup adds the GPIO23 passive piezo buzzer.

ML 1/5: model and camera-to-result workflow

The laptop detects visible people. A separate labeling workflow prepares future improvements.

Current model	YOLO11m, person class only. Confidence threshold 0.35, inference size 960. The reviewed laptop uses CPU-only PyTorch.
Camera input	Sensor agreement requests up to five JPEGs: 800 × 600. Originals are preserved and queued in SQLite.
Background inference	One worker loads the model once, analyzes each frame, saves results and reports its availability to Aegis.
Model output	Person verdict, bounding boxes, confidence, person count and inference time. Aegis groups frames into one event.

Person detection does not establish identity, authorization, intent or liveness.

ML 2/5: filtering and alarm decisions

A detection must pass image-level checks before it becomes an eligible alarm event.

Image preparation	Decode JPEGs and preserve proportions during resizing. Dark frames receive local contrast enhancement (CLAHE).
Accept a person box	Confidence ≥0.35, box area ≥0.5%, height ≥10% and width/height ≤3.0. Small or partial people can fail these checks.
Fallback and duplicates	If no box qualifies, run the same model with extra image padding. Merge overlapping boxes. TTA is off by default.
Alarm response	Sensor fusion requests a 1-second beep. The ESP32 polls for a fresh, new person event to request a 5-second ML alarm.

90-second freshness checks and event IDs suppress stale or repeated replies. Errors add no ML alarm.

ML 3/5: evaluation results and their limits

Recorded evaluations of person_ft.pt at threshold 0.50. The 218-image report is highlighted.

Report	Model / threshold	Images	Precision	Recall	False-positive rate
Historical	person_ft.pt / 0.50	227	98.56%	95.14%	2.41% (2/83)
Recovered	**person_ft.pt / 0.50**	**218**	**98.47%**	**94.85%**	**2.44% (2/82)**

These values describe saved person_ft.pt evaluations, not live YOLO11m performance. Training independence remains unverified.

ML 4/5: camera labels and dataset preparation

Human-reviewed originals provide the training truth for the next camera-specific model.

Current progress	792 original captures. Snapshot on 26 October: 325 saved labels, comprising 166 person images and 159 empty scenes.
Manual review	The local tool draws person boxes, confirms empty scenes and excludes uncertain images. Save labels with original-image hashes.
Difficult examples	Review dim light, blur, seated or partially visible people, unusual angles and blocked views. Quality scores guide review only.
Protected dataset split	Export reviewed labels into about 70% train, 15% validation and 15% test. Keep sessions and identical images in one split.

ML 5/5: fine-tuning, comparison and next steps

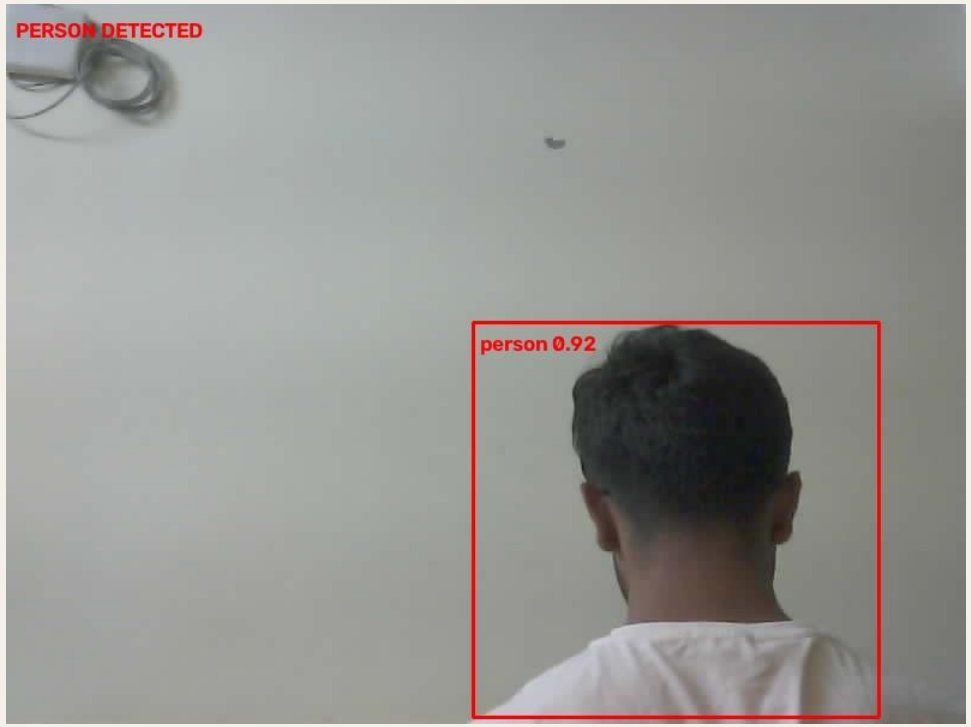
The tools are prepared. Camera dataset export, training and candidate deployment have not started.

Train a candidate	Start from YOLO11m. Up to 50 epochs, size 640, batch 2. Preserve the current model.
Tune on validation	Compare current and candidate models at thresholds 0.25, 0.35, 0.45 and 0.55 using the production detector and filters.
Test the whole event	Track missed people, precision, recall and empty-event false alarms. Fix the model and threshold before the final test.
Deploy after comparison	Review accuracy and latency, then update the model path and restart the worker. Keep rollback weights and validate the alarm response.

Camera evidence and confidence

Person detected

92.00% detection confidence



No person detected

No accepted person box



Limitations define the next improvements

Current limit	Next step
A person may fail the sensor gate.	Compare missed events across all three configurations.
Low light or a blocked view can hide a person.	Add a light. Aim at the entry path and keep the view clear.
Wi-Fi and camera failures can block verification.	Add explicit fault states, timeouts and an event queue.
Person detection does not establish authorization.	Add an armed schedule or access-control context.

Goal: fewer unnecessary alerts with acceptable missed-event risk.

How the project supports SDGs 9 and 11



Industry, Innovation and Infrastructure

Research and innovation

Combines dual sensing, camera verification and ML in an embedded security prototype.

Technical capability

Hardware integration and testing build skills in developing reliable monitoring systems.



Sustainable Cities and Communities

Safer shared spaces

Could help monitor building entrances and shared facilities for possible intrusions.

More informed response

Visual checks and saved events help users assess sensor triggers before taking action.

Alignment: SDG 9.5 and the safety aspect of SDG 11.7. Community safety impact still needs evaluation.

Live demo: software first, hardware second

01 Software demonstration

Open the event gallery and compare frames.

Show annotated boxes, scores and ML verdicts.

Explain a person and a no-person result.

Distinguish confidence from measured accuracy.

Website link

Open this link on the demo laptop while the local backend is running.

02 Connected hardware

Show PIR and ultrasonic readings together.

Enter the zone and observe camera capture.

Show the 1-second sensor beep and 5-second ML alarm.

Clear the zone and demonstrate rearming.

Research references and project evidence

[1] Pugh, Brewer & Kvam (2015)

Sensor Fusion for Intrusion Detection Under False Alarm Constraints. IEEE SAS. doi.org/10.1109/SAS.2015.7133634

[2] Ciuffreda, Casaccia & Revel (2023)

A Multi-Sensor Fusion Approach Based on PIR and Ultrasonic Sensors Installed on a Robot to Localise People in Indoor Environments. Sensors 23, 6963. doi.org/10.3390/s23156963

[3] Patino et al. (2022)

Fusion of Heterogenous Sensor Data in Border Surveillance. Sensors 22, 7351. doi.org/10.3390/s22197351

Project evidence: Velxio circuit, local evaluation logs and saved camera detections.

Full source URLs and presenter context appear in the speaker notes.